

COLD LAKE, AB, Canada

WMO: 711200

Lat: **54.420N**Lon: **110.280W**Elev: **1775**StdP: **13.78**Time Zone: **-7.00 (NAM)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.5	-22.0	-36.3	0.7	-27.2	-31.0	1.0	-21.8	24.5	27.2	20.7	21.9	3.0	230	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	82.5	64.5	79.5	62.9	76.7	61.4	66.7	79.2	64.8	76.1	63.0	73.4	8.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.0	88.7	72.1	60.3	83.6	70.1	58.8	79.1	67.9	32.3	79.3	30.8	76.1	29.4	73.6	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	18.1	15.9	DB	-34.1	88.6	6.0	3.5	-38.4	91.1	-41.9	93.1	-45.3	95.1	-49.6	97.6	(4)
(5)			WB	-34.3	70.3	5.9	1.8	-38.6	71.6	-42.0	72.6	-45.3	73.7	-49.6	75.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.3	5.6	10.2	22.5	38.4	50.9	59.2	64.0	61.3	52.0	39.0	21.7	8.5		
	DBStd	23.41	16.06	13.92	14.06	9.97	8.35	6.03	5.19	6.01	7.87	8.51	11.64	14.40		
	HDD50	6357	1375	1114	852	364	90	6	0	2	69	351	849	1285		
	HDD65	10591	1840	1534	1317	799	440	190	80	144	392	806	1299	1750		
	CDD50	1342	0	0	0	14	119	281	435	353	129	11	0	0		
	CDD65	103	0	0	0	0	4	16	50	30	3	0	0	0		
	CDH74	1247	0	0	0	4	99	202	507	356	77	2	0	0		
	CDH80	228	0	0	0	0	19	34	98	65	12	0	0	0		
(14) Wind	WSAvg	7.0	6.0	6.2	7.3	8.2	8.2	7.2	6.8	6.5	7.1	7.3	6.7	5.9		
(15) Precipitation	PrecAvg	18.3	0.8	0.6	0.7	0.9	1.7	2.8	3.5	2.9	1.8	0.7	0.9	0.9		
	PrecMax	25.7	1.7	1.6	2.1	2.5	4.0	7.2	7.3	6.7	4.0	2.0	2.4	2.4		
	PrecMin	13.0	0.4	0.1	0.0	0.1	0.2	0.3	1.6	0.3	0.4	0.2	0.1	0.2		
	PrecStd	3.5	0.3	0.4	0.5	0.5	1.0	1.8	1.4	1.5	1.1	0.5	0.5	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.8	43.1	53.1	72.5	82.5	84.0	86.8	86.2	81.7	69.3	49.5	41.4		
		MCWB	36.1	36.0	41.5	52.2	58.9	63.3	68.1	66.1	61.8	54.2	40.7	36.0		
	2%	DB	37.0	37.7	47.9	65.4	76.3	79.2	82.7	81.2	75.5	61.4	43.7	36.0		
		MCWB	33.2	32.6	38.8	48.0	56.5	61.0	66.1	64.5	59.6	49.2	37.6	32.1		
	5%	DB	32.8	33.3	43.8	59.8	71.7	75.7	79.5	78.0	71.0	56.4	39.7	32.1		
		MCWB	29.6	29.7	36.4	45.1	53.6	59.3	64.4	63.0	57.3	46.2	35.4	29.3		
	10%	DB	28.1	29.8	40.3	55.0	67.4	72.2	76.4	74.7	66.4	52.3	36.1	28.2		
		MCWB	25.7	27.0	34.2	42.9	51.5	57.3	62.7	61.5	54.8	44.0	32.4	26.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.8	36.7	42.8	53.1	60.6	66.0	70.0	68.4	63.6	55.1	41.8	36.4		
		MCDB	41.1	42.1	52.0	69.7	76.7	78.8	83.3	82.0	77.5	67.7	47.6	40.6		
	2%	WB	33.5	32.8	39.2	48.9	57.7	63.3	67.7	65.9	60.5	50.4	38.6	32.4		
		MCDB	37.1	36.7	46.9	63.9	73.1	74.9	80.5	78.4	73.4	59.1	42.9	35.4		
	5%	WB	29.6	30.0	36.8	45.9	55.1	61.2	65.8	64.2	58.0	47.2	35.6	29.4		
		MCDB	32.5	33.2	43.2	58.6	68.6	71.6	77.0	75.3	69.3	55.3	39.3	32.0		
	10%	WB	25.7	27.0	34.6	43.3	52.9	59.3	63.9	62.4	55.6	44.3	32.8	26.0		
		MCDB	28.1	29.8	39.8	53.9	64.8	69.0	73.8	72.5	64.7	51.3	35.9	28.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.7	18.3	18.6	19.4	21.7	19.7	19.8	20.2	20.0	16.1	13.0	14.1		
		MCDBR	19.0	22.1	21.2	28.0	28.8	25.4	24.7	26.2	28.5	24.8	17.1	17.7		
		MCWBR	15.9	17.9	14.1	15.6	13.9	11.3	11.5	12.4	15.1	15.1	12.4	15.1		
	5% WB	MCDBR	18.8	21.3	20.6	26.3	25.5	21.3	22.3	23.0	26.5	23.5	16.6	17.2		
		MCWBR	16.0	17.6	14.0	15.2	12.9	10.8	11.0	11.6	14.5	14.8	12.6	14.9		
(40) Clear-Sky Solar Irradiance	taub		0.246	0.275	0.279	0.328	0.346	0.364	0.363	0.367	0.312	0.278	0.248	0.218		
	taud		2.305	2.259	2.393	2.397	2.397	2.356	2.409	2.355	2.516	2.582	2.459	2.340		
	Ebn at Noon		232	260	289	286	285	280	278	269	275	264	235	224		
	Edh at Noon		17	26	28	33	35	36	34	34	25	19	15	13		
(44) All-Sky Solar Radiation	RadAvg		272	601	1087	1477	1799	1834	1866	1521	1066	595	273	190		
	RadStd		28	48	76	121	154	129	105	121	116	97	36	18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page